33.3 A 125-to-170GHz Power-Efficient Phase Shifter in SiGe BiCMOS with Outphasing Gain and Phase Corrections

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Advancements in silicon technologies are opening the way to sub-THz phased-array transceivers, enabling high-resolution radar sensors, and wireless communications with a fiber-like transport capacity.

Programmable phase shifters (PSs), not yet deeply investigated, are crucial components in such systems. At the RX side, the PS follows a low-noise amplifier (LNA), with both low noise and high linearity being key to maintain the dynamic range. Conversely, on the TX side, the PS drives a power amplifier (PA), and high output power at 1dB gain compression (OP_{1dB}) for the PS is highly desirable to relax PA requirements and improve system efficiency.

Most of the PSs in D-band rely on the vector-sum principle, where signals in quadrature are weighted by variable-gain amplifiers (VGAs) and combined to achieve the desired phase shift [1-3]. This architecture grants fine phase resolution, but the need for VGAs with a wide range of gain-control compromises the linearity, limiting the OP_{1dB}. Linear PSs based on passive components have also been investigated. [4] presented a 140GHz PS based on a digitally tapped delay line. The remarkably high OP_{1dB}=3.4dBm comes with 10dB attenuation and a narrow bandwidth (10GHz only), limited by the parasitic capacitances of the CMOS switches along the line. [5] proved broadband operation by routing the signal through transmission lines of different lengths, at the expense of a 20dB insertion loss, coarse phase control, coverage of 0° to 360° only in the lower D-band portion, and large chip size.

This paper presents the hybrid (passive/active) vector-sum PS architecture in Fig. 33.3.1, comprising wideband programmable passive networks and amplifiers that compensate for the insertion loss. The architecture removes VGAs for weighting the vectors to be summed and replaces them with two amplifiers that are biased at the optimal condition for maximum OP_{1dB} with constant gain. Experimental results on a SiGe BiCMOS test chip demonstrate wideband operation (125 to 170GHz) with 0-to-360° phase control in 9° steps. Compared to active vector-sum PSs in the D-band, measurements show comparable gain, phase resolution, noise figure (NF), and power consumption (P_{DC}) but with a higher OP_{1dB}, leading to 5× enhancement of the power efficiency ($\eta = OP_{1dB}/P_{DC}$).

The operation principle of the proposed PS is explained by Fig. 33.3.1. Looking at the block diagram (top-left), the input signal S_{in} is split in two quadrature paths (I/Q). Then, the two quadrature signals feed a pair of blocks, referred to as $\Delta\varphi_{I,Q}$, which introduce a programmable phase shift in a range of (at least) 0° to 90° with fine-enough resolution. Assuming the phase shifters are controlled by the same digital word ($\Delta\varphi_I=\Delta\varphi_Q=\Delta\varphi$), the outputs on the I and Q paths are $(S_{in}/\sqrt{2})e^{j\Delta\varphi}$ and $j(S_{in}/\sqrt{2})e^{j\Delta\varphi}$, respectively. The two components are then fed to 0°/180° phase shifters controlled by 1b signals, sw_I, sw_Q, and are eventually summed at the output of two identical amplifiers (with gains G). Assuming lossless passive networks, the overall PS gain, $G_{PS}=S_{out}/S_{in}$ and phase ($\angle G_{PS}$) are summarized in the table of Fig. 33.3.1, showing that with $\Delta\varphi \in$ [0° to 90°] and a proper combination of sw_I, sw_Q, the input-output phase shift spans the 0°-to-360° range, with the resolution set by $\Delta\varphi$. It is worth noticing that in each path the phase shift is programmable across two quadrants only (0° to 90° and 180° to 270°), limiting the implementation complexity of the required networks. The insertion loss of the passive networks is nevertheless dependent on the phase settings, thus programmable loss compensation is needed to produce an output with a variable phase but constant amplitude. To avoid VGAs, regulation of the gain is performed by independently controlling the phase shifts in the I/Q paths $\Delta\varphi_I$, $\Delta\varphi_Q$. The concept, illustrated in Fig. 33.3.1 (top-right), is the same as that exploited in outphasing amplifiers [6]. The two quadrature signals produce an output with phase φ_{out}. By properly varying the relative phase between the two vectors ($\Delta\varphi_I=\Delta\varphi-\delta\varphi$, $\Delta\varphi_Q=\Delta\varphi+\delta\varphi$), it is possible to change the amplitude by δG but not φ_{out}. The independent control of $\Delta\varphi_I$, $\Delta\varphi_Q$ is also leveraged to correct for deviations from quadrature of the two vectors being summed, due for example to a phase error of the I/Q input splitter or deviations from the 0°/180° of the phase-inversion blocks. The situation is depicted in Fig. 33.3.1 (bottom-right). The quadrature error $\delta\varphi_{err}$ leads to an error $\delta\varphi$ on φ_{out}, nulled by $\Delta\varphi_I=\Delta\varphi$ and $\Delta\varphi_Q=\Delta\varphi-\delta\varphi_{err}$.

The detailed schematic of the realized PS is drawn in Fig. 33.3.2. The input splitter is a quadrature hybrid, realized with coupled lines of 245µm length, corresponding to λ/4 at 150GHz. The programmable phase shifters, $\Delta\varphi_I$, $\Delta\varphi_Q$, are implemented by cascading four band-pass filters with center frequency tunable by a digitally switched capacitor bank. The programmable range of the phase shift is purposely selected greater than 90° to support a rotation of the output vector over one quadrant with a margin to be used for gain and phase corrections. The simulated input-output phase shift (Fig. 33.3.3, left) is programmable over a range of 130° at 125GHz and 166° at 170GHz in 17 steps with a thermometric control. The insertion loss spans 5 to 9.5dB in the 125-to-170GHz band. The networks for phase inversion (0°/180°) are realized as two-state reflective-type phase shifters using quadrature hybrids. The signals $S_{A,I/Q}$ are equally split at the CPL/DIR ports, terminated by the NMOS switches controlled by sw_I, sw_Q, with reflection coefficients $\Gamma_{SW,on/off}$ ideally ±1 ($\Gamma_{SW,on/off}=(Z_{SW,on/off}-Z_0)/(Z_{SW,on/off}+Z_0)$, with Z_0 the characteristic impedance of the hybrid and $Z_{SW,on/off}$ the on- and off-state impedance of the switches). The signals reflected by the switches sum at the ISO port, making the outputs $S_{B,I/Q}$ with a gain magnitude and a relative phase shift equal to $\Gamma_{SW,on/off}$. Inductors (L_{sw}) shunt the NMOS switches to resonate out the parasitic capacitance, and the transistors are sized (W/L=50µm/55nm) to achieve $|\Gamma_{SW,on}|\approx|\Gamma_{SW,off}|$ at 150GHz, giving the same network insertion loss in the on and off states. $|\Gamma_{SW,on/off}|\approx0.7$ results in ≈3dB loss, which sums to 0.8dB loss of the couplers. The simulated phase shift across frequency (Fig. 33.3.3, right) ranges from 146° at 125GHz to 211° at 170GHz. The relatively large phase deviation from 180° at the band edges is finally compensated by properly selecting $\Delta\varphi_I$, $\Delta\varphi_Q$. The signals $S_{B,I/Q}$ feed a pair of cascode amplifiers with input-matching networks MN_1. All the HBTs, (8µm×0.2µm emitter area) are biased with ≈8mA from a 2V supply. The collectors of Q_2 are shorted to sum the I/Q paths and inject their currents into the shared network MN_2, which performs a step-up transformation of the 50Ω off-chip termination to raise the gain.

The PS was realized in a SiGe BiCMOS 55nm technology with a footprint of 800×250µm². The chip was wire-bonded on a PCB that provides biasing, supply, and digital signals to an on-chip serial interface. S-parameters measurements were acquired using an Agilent E8361C VNA coupled to VDI WR6.5-VNAX extension modules and Infinity waveguide 75µm-pitch GSG probes. The setup was de-embedded up to the probe tips using an external TRL cal kit. Measurement results (S_{21}, S_{11}) in all possible PS configurations are shown as grey curves in Fig. 33.3.4. An offline calibration was applied to select the configurations giving the minimum rms phase error from the ideal curves over a target bandwidth, with a maximum gain variation within ±1dB. The procedure was applied within three sub-bands, centered at 138GHz, 152GHz, and 165GHz, to reduce the rms errors across a larger bandwidth. The selected S_{21} curves are drawn in red, yellow, and blue colors in Fig. 33.3.4.

Figure 33.3.5 (top) shows the polar plot (with the magnitude in a linear scale) at the three center frequencies. The bottom plot in Fig. 33.3.5 shows the rms magnitude and phase errors over frequency, which are within 0.8dB and 5° respectively, across the 125-to-170GHz band. The simulated noise figure, extracted at center frequency (150GHz) across the 0°-to-360° phase shift, spans 16 to 18dB.

Large-signal measurements were obtained using an ELVA-1 DPM-06 D-band power meter. Figure 33.3.6 shows the measured OP_{1dB} (top-left) and efficiency (top-right) at three frequency points for a 0°-to-360° phase shift. The OP_{1dB} is always greater than 2dBm and, with power consumption nearly constant (P_{DC}=31mW), the power efficiency, $\eta=OP_{1dB}/P_{DC}$, ranges from 5.2% to 7.1%. The measured results are finally summarized in the table of Fig. 33.3.6 and compared against previous works in similar frequency bands. Fully passive PSs are narrowband [4] or display excessive insertion loss and die size [5]. Compared to the active vector-sum PSs, the presented chip shows comparable or better performance on most of the aspects. By removing the VGAs in the proposed architecture, the PS achieves an OP_{1dB} significantly higher than in previous works, and when normalized to the power consumption gives the highest reported efficiency, at least 5× better than other PSs working above 100GHz in Fig. 33.3.6. The die micrograph is shown in Fig. 33.3.7.

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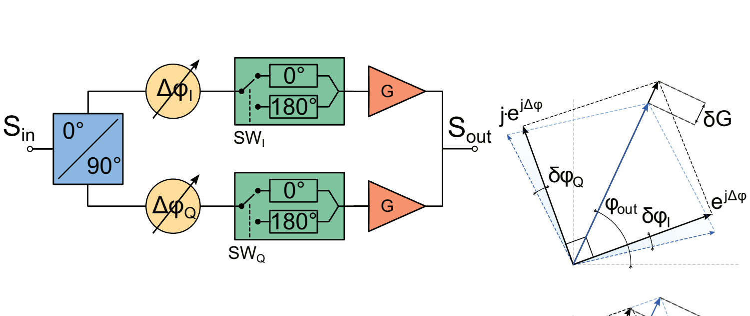


$SW_{I/Q}$	G_{PS}	$\angle G_{PS}$
0/0	$\frac{G}{\sqrt{2}}\cdot(1+j)\cdot e^{j\Delta\varphi}$	[45° to 135°]
0/1	$\frac{G}{\sqrt{2}}\cdot(1-j)\cdot e^{j\Delta\varphi}$	[315° to 45°]
1/0	$\frac{G}{\sqrt{2}}\cdot(-1+j)\cdot e^{j\Delta\varphi}$	[135° to 225°]
1/1	$\frac{G}{\sqrt{2}}\cdot(-1-j)\cdot e^{j\Delta\varphi}$	[225° to 315°]

Figure 33.3.1: Block diagram and operation principle of the phase shifter.

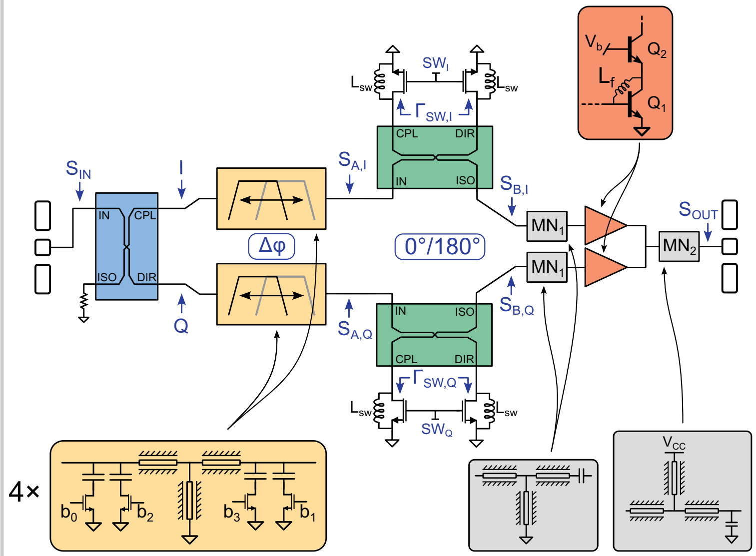


Figure 33.3.2: Schematic of the implemented phase shifter.

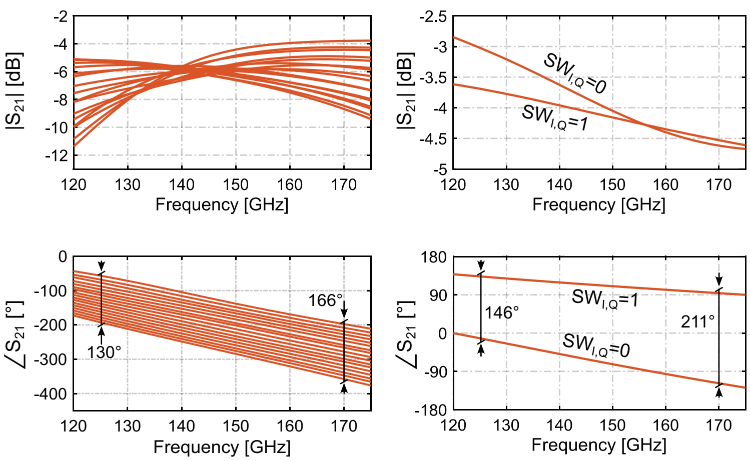


Figure 33.3.3: Simulated insertion gain and phase of the fine-step tunable tanks (left) and reflective-type phase inverter (right).

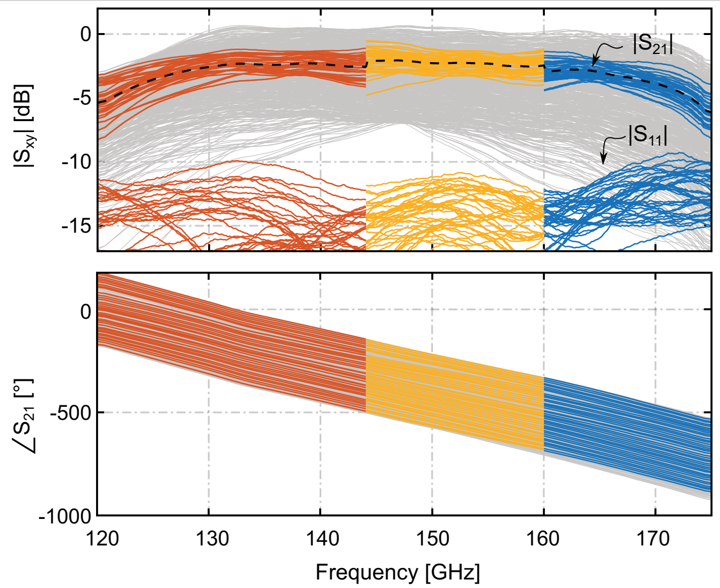


Figure 33.3.4: Measured S-parameters. The curves selected after calibration are highlighted.

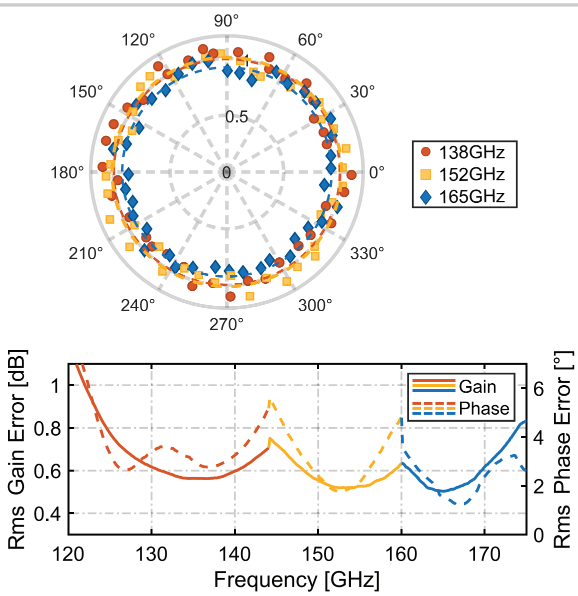
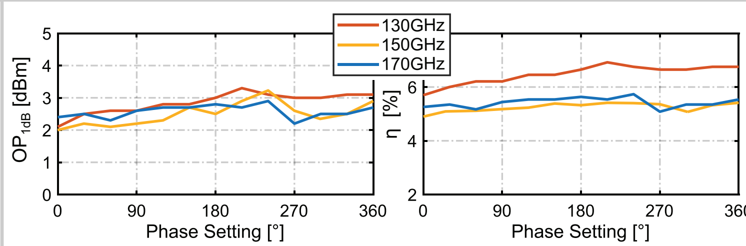


Figure 33.3.5: Polar plot at the calibration frequencies (top) and corresponding rms phase and amplitude errors across frequency (bottom).



Reference	**This Work**	[1]	[2]	[3]	[7]	[8]	[4]	[5]
Topology	Vector-Sum						Passive	
Technology	55nm SiGe	55nm SiGe	90nm SiGe	55nm SiGe	0.13μm SiGe	0.13μm SiGe	45nm RFSOI	0.13μm SiGe
Core Area [mm^2]	0.20	0.81	0.02	0.05	-	0.04	0.04	1.16
Frequency [GHz]	125 to 170	110 to 145	150 to 210	140 to 160	78 to 105	85 to 100	135 to 145	110 to 170
Gain [dB]	-2.3	1.5	-8.7	-4.5	2	-7.8	-10.5	-22
Phase Res. [°]	9	-	90	11.25	5.6	5.6	11.25	17 to 27
RMS ϕ_{err} [°]	5	8.5	16	7.5	3	3.4	1.2	4
RMS $Gain_{err}$ [dB]	0.8	1.2	1.5	1.4	0.9	0.7	0.5	1
NF [dB]	16 to 18§	-	-	19 to 20.5§	12	-	-	-
OP_{1dB} [dBm]	2 to 3.5	-6	-15.4	-3.7	0	-10.3	3.4	-
P_{DC} [mW]	31	20.3†	5.4	50†	49.5	62	0	6.22
η @ P_{1dB} [%]	**5.1 to 7.2**	1.2†	0.5†	0.8†	2	0.15	-	-

§ Simulated value † Average value on the constant gain circle

Figure 33.3.6: Measured OP1dB and efficiency at three frequencies vs. phase settings (top). Performance summary and comparison (bottom).

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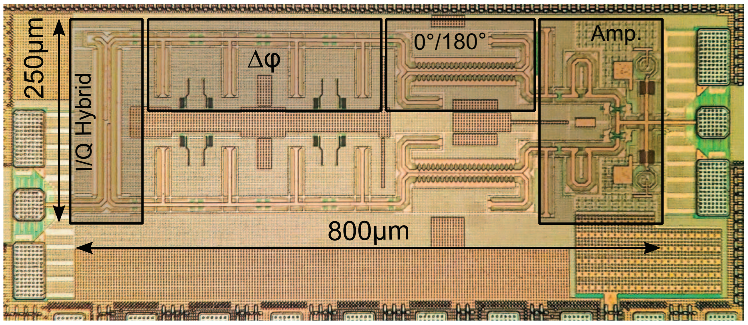

Figure 33.3.7: Die micrograph.

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